← Back **Quiz 1: History and Evolution of SDN**
Graded Assignment • 30 min

Due Mar 11, 11:59 PM CET

1. Which of the following constituted advantages to AT&T's elimination of in-band signaling from the phone network in the 1980s? 1 point

- [ ] Easier configuration and management
- [ ] Improved security
- [x] Introduction of new services
- [x] Reduction of expenditures

2. What types of functions were enabled by active networks? 1 point

- [x] Code installed on nodes in the network (e.g., at routers and switches) could process packets.
- [ ] Code running on end hosts (e.g., machines sending and receiving packets) could modify packet contents.
- [ ] All routers in the network needed to support active networks for active networks to be deployable.
- [x] A single node in the network could perform more than one function on a packet or stream of packets (e.g., firewalling, followed by load balance).
- [x] Packets carrying code in the payload could be executed on intermediate nodes in the network (e.g., routers, switches)

3. Which of the following is true about network virtualization? 1 point

- [ ] Network virtualization was invented in the late 2000s to support the migration of virtual machines in data centers.
- [x] Network virtualization allows multiple service providers to share the same underlying physical infrastructure.
- [ ] Network virtualization requires the instantiation of virtual machines at each intermediate node (e.g., switch, router) in the network.
- [ ] Network virtualization requires every router and switch in the network to run custom technology supporting virtualization.
- [x] Virtual networks can be realized by dividing switch forwarding tables based on fields in the packet header (e.g., IP address and port).

4. Which of the following are true about various systems developed for central control of packet-switched networks? 1 point

- [x] Ethane's initial goal was to support authentication of hosts in enterprise networks.
- [ ] The Routing Control Platform requires changes to existing routers to support central control.
- [ ] The OpenFlow protocol requires vendors to expose the details of how they implement flow table maintenance in their switches.
- [x] The RCP could be deployed within a single ISP network, without coordination across multiple ISPs.
- [ ] The FORCES protocol is deployed on most major routers and switches today.

5. Which three artifacts and lessons that active networks affected the development of software-defined networks? Select all that apply. 1 point

- [x] packet headers
- [ ] distributed state management
- [x] demultiplexing
- [ ] centralized control
- [x] programmable functions